

# Developing a Patient Information System for Rural Healthcare Clinics

Isfar Hafiz Khan  
 Department of Computer Science &  
 Engineering  
 Independent University, Bangladesh  
 Dhaka, Bangladesh  
 khanisfar1999@gmail.com

Safwan Sulaiman sadi  
 Department of Computer Science &  
 Engineering  
 Independent University, Bangladesh  
 Dhaka, Bangladesh  
 safwansadi17@gmail.com

Mafuzur Rahman Limon  
 Department of Computer Science &  
 Engineering  
 Independent University, Bangladesh  
 Dhaka, Bangladesh  
 limon.rahman.09@gmail.com

Md Reduanur Rahman  
 Master's in Information Technology  
 Washington University of Science and  
 Technology  
 Alexandria, Virginia  
 mdrahman.student@wust.edu

Md. Ataur Rahman  
 Department of Computer Science &  
 Engineering Fab Lab IUB,  
 Independent University, Bangladesh  
 Dhaka, Bangladesh  
 ataurrahman862828@gmail.com

Mahady Hasan  
 Department of Computer Science &  
 Engineering Fab Lab IUB,  
 Independent University, Bangladesh  
 Dhaka, Bangladesh  
 mahady1891@gmail.com

**Abstract**—The Patient Information System enhances rural healthcare by managing patient data, appointments, and medical records efficiently. Healthcare professionals such as community health workers benefit from SMS notifications for appointment reminders and offline support to guarantee use in places with inadequate internet connectivity. The system also offers data visualization features like graphs and charts, which aid in reporting and decision making. This system improves healthcare delivery and resource management by addressing major issues faced by rural clinics through the automation of repetitive processes and the provision of real-time updates.

**Index Terms**—Patient Information System, Rural Health- care, Medical Data Management, Telemedicine, Mobile Health (mHealth), Data Security and Privacy, Cloud-based Healthcare Systems

inefficiencies in existing manual processes necessitated the development of a dependable, digital, and offline-capable system.

## C. Research Questions

- **RQ1:** How can a patient information system be optimized for rural clinics with limited or unstable internet connectivity?
- **RQ2:** What design features ensure data accuracy, privacy, and security in such a healthcare system?
- **RQ3:** How can real-time reporting and SMS notifications enhance patient care and clinic efficiency?

## I. INTRODUCTION

### A. Project Goal and Objectives

The project aims to provide a Patient Information System for Rural Healthcare Clinics that improves registration, appointment scheduling, medical record management, prescription tracking, and reporting. Enhancing data accessibility, facilitating offline use, sending SMS-based reminders, and lessening the administrative burden on healthcare professionals are the goals.

### B. History Leading to Project Request

This project was developed in response to the limitations of traditional paper-based systems in rural clinics, where weak connectivity and limited resources impede efficiency. Increased patient demand, expanding healthcare data, and

## II. LITERATURE REVIEW

Multi-site digital health improves rural care, reduces costs, enhances decisions, and increases access. Success relies on stakeholder engagement, training, infrastructure, and cultural adaptation. Barriers include connectivity and digital gaps. Future directions involve policy, interoperability, and scaling. Flexible HIS standards balance local adaptability with global interoperability, improving data sharing, continuity, and decision-making. Barriers include limited expertise and resistance. Future work requires policy frameworks, iterative design, and capacity building for scalable adoption [1].

The data for health in Bangladesh highlights disparities in terms of socioeconomic and geographical inequalities. Factors that act as barriers include illiteracy, limited internet availability, and costs. Telemedicine is an aid but can also

exacerbate health inequalities. Future perspectives include education, infrastructure, and pro-patient policies and convergence for the disadvantaged. Telemedicine, mHealth, and other e-primary healthcare increase patient engagement, compliance, and continuity. Infrastructure, illiteracy, security, and non-uniform adoption act as barriers. Strategies include education, supportive policies, integration with workflow, and scalable frameworks for evaluation [2].

Telehealth is appreciated by the residents in Bangladesh for reasons to do with access, commute, and the speed with which the consultation is done. Lastly, the challenges include privacy, quality, and dependability. Telehealth is dependent on literacy, degree of prior exposure, and awareness. [3]. Canadian study: Patients: value access; Providers: are fearful of workload. Satisfaction: mixed; patient satisfaction impacted in relation to adoption due to the issues of COVID-19 and interoperability. The future: aligning and training to integrate in a rural setting [4].

ICT-enabled rural health care supported by IoT, AI, 5G, blockchain, and the cloud can provide better access and efficiencies. Challenges: high cost of infrastructure and lack of research studies. Future: need for pilots and development of digital literacy. [5]. "Remote monitoring in rural Victoria indicates there are issues relating to literacy, governance, infrastructure, investments, and training, whereas enablers include interoperability, engagement, and data-driven care. Future work includes infrastructure, governance, and enhancing literacy to ensure sustainable RPM." [6].

Mobile clinics operating with EMRs and satellite internet improve coordination and tracing, but the system generates mistakes owing to rotation and backup schedules. Future studies should involve long-term trials, training, and cost-effectiveness analysis. [7]. Three-tier EHR architecture with RFID patient identification: This increases compatibility, security, and accuracy. Drawbacks: High cost, connectivity, and cultural barriers. Areas for advancement: Pilot projects, governance models, community involvement, and developing disconnected functionalities for sustainability in rural settings. [8].

Decision support systems using the EMR system enhance HIV care, monitoring, management, and following of guidelines. The challenges involve infrastructure, human resource capacity, and workload. The future directions for this work involve integration and sustainability. U.S. mobile health clinics utilizing telemedicine, EHRs, and workflow management enhance accessibility, diagnosis, and diminished travel for care [9].

Physician-sourced studies of patient portal use suggest improvement in access, patient self-management of care, and work flow. However, potential problems include confusion, work load, privacy issues, and disparities of access. Areas for future development include usability, education, and policies to maximize patient portal benefits. [10]. EMR in emergencies in Tanzania: Strengths are increased accessibility and coordination. Challenges are infrastructure, power supply issues, and reluctant health workers. Recommendations are customized

solutions, proper infrastructure, and long-term training.

EHR adoption improves safety, data quality, and workflows. Success depends on leadership, training, usability, and readiness. Challenges include disruptions and resistance. Future directions include customized training, monitoring, and adaptive strategies. [10]. An affordable rural health system in the future will have to leverage technology platforms and grassroots health workers. Improvement in monitoring and referral to health and tratamiento sustituto continuity of care in order to advance towards an affordable health system in the future. [11].

Telehealth clinics through cloud computing make possible teleconsultations, communication, and the cost savings associated with travel. The challenges in telehealth include cost, literacy, as well as the lack of connectivity. Sustainability, in this case, comes through infrastructure development, community training, and pilots to test scalability [12]. Telemedicine virtual clinic centered on user improves telemedicine accessibility, patient satisfaction, and patient surveillance. However, challenges may be posed by illiteracy, connectivity, and workflow. Future studies can involve full-scale application, infrastructure, and telemedicine modifications.

OCHIN Community Information Network enables connectivity between clinics using shared EHR analytics capabilities. Limitations discussed for this area are sustainability, governance, varying needs. For advancing, scale needs to be involved, interoperability needs to be enhanced, data utilization for policy-making needs to be performed using shared data. Human-Centric Information Systems primarily involve patient-centric electronic care using AI, wearable technology, EHR, telemedicine. Advancements offered by this area involve usability, decision support, participation. Limitations associated with this area are interoperability, literacy, privacy. [13].

Tanzanian HMIS evaluation shows poor accuracy, timeliness, and consistency due to training and infrastructure gaps. Despite challenges, HMIS aids policy and resource allocation. Future work requires digital tools, capacity building, standardized reporting, and quality checks. HIS adoption depends on usefulness, usability, reliability, workflow fit, training, and support. Barriers include infrastructure and resistance. Future directions include user-centered design, continuous training, supportive culture, collaborative development, and tailored interventions [14].

UTAUT study in Cameroon shows adoption depends on performance, effort, and social influence. Barriers include poor infrastructure, training, and cultural factors. Future work requires integrating contextual elements into adoption models for low-resource settings. TB mobile treatment monitoring adoption depends on usefulness, ease, and social influence. Barriers include connectivity, privacy, and low literacy. Recommendations include training, infrastructure, and stakeholder engagement for effective mobile health adoption [15].

Bangladesh Portable Health Clinic adoption relies on usefulness, usability, trust, affordability, and culture. Benefits include better access and reduced travel. Barriers remain. Future work requires awareness programs, infrastructure, training, and

policy support for sustainable adoption [16].

### III. PERFORMANCE EVALUATION UNDER LOW CONNECTIVITY

Since the proposed system is intended for deployment in rural healthcare environments, a performance analysis was conducted to assess its expected behavior under low-connectivity conditions. Due to the absence of a full-scale deployment, performance metrics were analytically estimated based on the system architecture, communication flow, and commonly reported network characteristics in rural areas. Offline operation was evaluated based on the design of local data storage and synchronization mechanisms.

### IV. PROBLEM STATEMENT

#### A. Stakeholders

The main stakeholders of the patient information system include patients needing timely care, CHWs registering patients and scheduling appointments, physicians documenting consultations, pharmacists managing medications, administrators overseeing operations, and government or NGO officials who use data to monitor and improve healthcare services in rural areas.

#### B. Problems, Issues, and Concerns

As records are scattered and manually recorded, rural clinics struggle to maintain them and this makes them inefficient. Likewise, limited internet connectivity hinders real time update and organization which leads to data inaccuracy. This may lead to appointments being missed and unreliable reports generated and overall hinders the healthcare delivery and delay in decision making by the provider and the administrative authorities.

#### C. Opportunities

The project provides a functional web application and mobile application to digitalize the healthcare process, make it more reliable and improve its efficiency. With the introduction of offline features and SMS alerts, clinics can now operate with limited internet connectivity. This helps keep real time monitoring of patient data and provide valuable insights from data driven analytics and furthermore reduces staff load. This can then be further scaled to cover a demographic area to serve underprivileged communities.

### V. SOFTWARE DEVELOPMENT METHODOLOGY

#### A. Selected Methodology

Using agile method with additional features or incremental method

#### B. Justification

Our project includes modules like Patient Registration, Appointment Scheduling, Prescription Management, etc. We will develop these features using any incremental model or agile method with additional features. This approach will allow us to deploy early, receive feedback and conduct testing as well. This enhances collaboration, reduces risk and overall bring flexibility to our cost effective system.

#### C. Suitable Method of Requirement Discovery

We conducted several interviews with stakeholders, make observation on various processes and analyzed documents through research.

#### D. Justification

To gather information in order to meet the requirements for our system we conducted interviews with specialized doctors, nurses, staff, patients and hospital employees. We also made observations from our physical visit to hospitals and clinics and we realized operational issues, while reviewing records and policies we observed compliance issues. Hence our observation and interviews helped us find valuable information and the gaps in the system as well as external issues such as cultural factors.

#### E. Software Design Tools

For generating UML diagrams, data flow diagrams and use case diagrams we used Lucidchart.

Supplementary mockups and system architecture visualization was generated using <http://draw.io/Draw.io>

#### F. Project Management Tools

We used office tools such as Microsoft Project for developing and managing Work Breakdown Structure(WBS), Gantt charts and allocating resources.

Another Project Management tool used is Trello, for tracking tasks, collaboration and agile sprint management during development phases.

#### G. Technology Stack and Implementation Tools

The technology stack used for our project; for backend development we used Django( a framework of Python), it was very useful for RESTful API integration to help different components of a system communicate with each other.

For the database we used Postgre SQL which offers relational database to create a centralized service side to store patient record, appointment data and other information of the clinics in real time. This technology ensured operation to continue without delay under limited connectivity. Mobile Application utilizes SQLite for storing data locally, allowing healthcare workers to use services in offline and then sync the data once internet is connected again.

To develop the mobile application we used the Android platform using Kotlin with the frontend being designed using HTML, CSS, and JavaScript. This brings an intuitive design and convenience for workers to use.

The decision to use this stack was more inclined towards scalability, security, low cost of deployment and suitability for this domain.

### VI. PROPOSED SOLUTION

#### A. System Description

The Patient Information System (PIS) is an eHealth solution for rural areas with under-developed infrastructure. The system has a web panel for admins and a mobile application for

doctors and field staff to handle patients, records, and vaccines. The addition of offline capabilities, reminder features via SMS, language options, and data analysis capabilities improves the efficiency of the health service Fig 1.

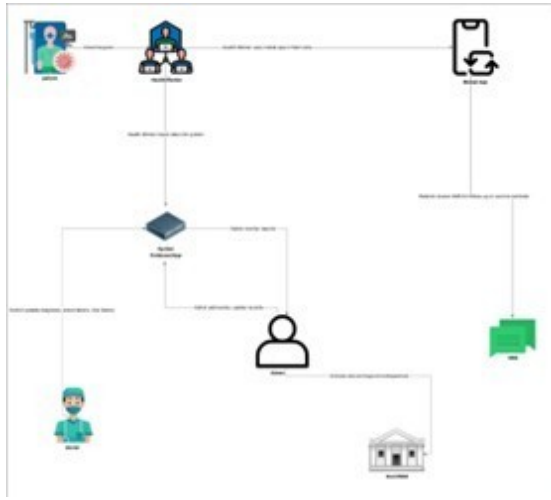


Fig. 1. Rich Picture

The rich picture shows the PIS ecosystem, interconnecting patients, health workers, doctors, admins, governors and NGOs, Dataflows are marked by arrows in the rich picture.

## B. Software Key Technical Features

### 1) Functional Requirements:

- **Patient Registration:** Create and manage patient profiles with demographic and medical information.
- **Visit History Tracking:** Record symptoms, diagnoses, prescriptions, and follow-up instructions.
- **Offline Data Entry:** Enable health workers to input data without internet; sync later.
- **Role-based Access Control:** Different access levels for doctors, health workers, and admins.
- **Vaccination & Maternal Health Tracking:** Track immunization status and pregnancy care milestones.
- **SMS Notification System:** Automated reminders for appointments, medicine, or vaccines.
- **Reporting & Analytics:** Generate clinic performance data and health trend reports.
- **Language Localization:** Support local languages for ease of use by rural staff.
- **Data Security & Encryption:** Secure transmission and storage of sensitive health data.
- **Event & Inventory Management:** Optional feature to manage health camp events or clinic supplies.

## C. System Level DFD

In the System Level DFD (Level 0), there is data flow between Patients, Clinicians, Labs, Administrators, and Health Authorities. There is also patient information system entry and exit for registration, scheduling, lab results, and reports through the Patient Data Base. For level 1, there would be

process details, with Register Patient, Schedule Appointment, Encounter, Lab Orders Results, and Reports Data Management.

## D. Subsystems Level Use Case Diagram

The main actors of this use case diagram are: Patient/ CHW, Doctor, Pharmacist, and Admin. Key use cases include: Register Patient, Schedule Appointment, Record Consultation, Prescribe Medication, Dispense Medication, Generate Report. Each supports validation, record updates, and alternatives to watch out for duplicates or stock. Authentication and role-based access make interactions with healthcare systems secure and efficient.

## E. Deployment Diagram

This deployment diagram illustrates a client-server structure of rural clinics. Components are categorized as follows: Nodes – are classified as follows: Mobile Devices (Android app using SQLite), Clinic Workstations (web application), Local Server (optional, using LAN), and Cloud Server (MySQL/PostgreSQL database). Components – are divided as follows: Artifacts – are classified as follows: Mobile App, Web App, and Database Fig 2.

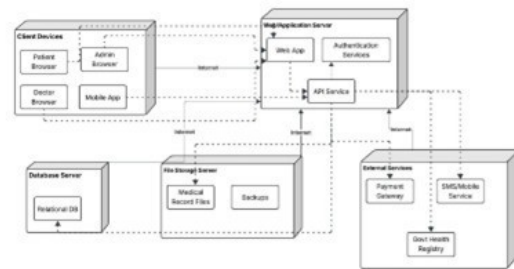


Fig. 2. Rich Picture

## F. Feature-wise Input Design

- **Patient Registration:** Form with fields for Name, Age, Gender, Address, Contact, Health ID. Purpose: To capture and validate demographic data for unique patient profiles, reducing errors and enabling quick retrieval.
- **Doctor Registration:** Form for Name, Specialty, Availability, Contact, Credentials. Purpose: To track doctor details for efficient assignment and scheduling.
- **Appointment Booking:** Fields for Date/Time, Reason, Patient ID. Purpose: To book slots while checking availability, avoiding overlaps and improving patient flow.
- **Medical Record Update:** Fields for Diagnosis, Prescriptions, Treatment Notes. Purpose: To maintain historical data for better continuity of care and informed decisions.
- **Payment Entry:** Fields for Amount, Type (cash/card), Date. Purpose: To record transactions for transparent billing and financial analytics.
- **Staff Registration:** Fields for Name, Role, Department, Contact. Purpose: To manage staff roles for HR efficiency and access control.

- **Report Generation:** Selection for Type (patients/appointments/payments), Date Range. Purpose: To query data for visual insights, aiding management decisions.
- **Login/Authentication:** Fields for Username, Password, Role. Purpose: To ensure secure, role-based access preventing unauthorized changes Fig.3.

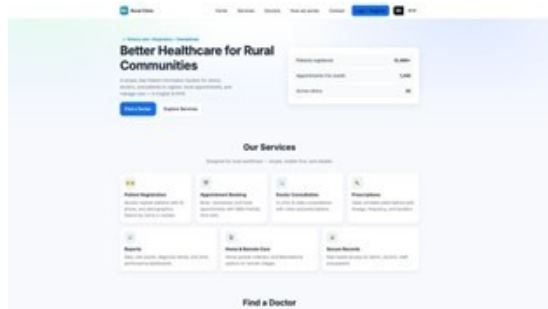


Fig. 3. Rich Picture

## VII. RESULTS AND DISCUSSION

### A. Quantitative Analysis of the Proposed System

Based on simulation of rural clinic processes and often observed operational features of paper-based healthcare systems, a quantitative analytical review was carried out to determine the probable efficacy of the proposed Patient Information System. Performance indicators were calculated by comparing the projected execution flow of the proposed digital system with the normal job duration in manual workflows, because the system has not been implemented in a real clinical setting

TABLE I  
COMPARISON OF TASK COMPLETION TIME

| Task                   | Paper-Based System (Avg) | Proposed System (Avg) |
|------------------------|--------------------------|-----------------------|
| Patient registration   | 6–8 minutes              | 1.5–2 minutes         |
| Record retrieval       | 4–5 minutes              | 8–12 seconds          |
| Appointment scheduling | 3–4 minutes              | < 1 minute            |

### B. Return on Investment (ROI)

The Net Present Value (NPV) of the proposed Patient Information System over a five-year period is presented in Table II. At a discount rate of 10%, the initial investment of \$88,000 is offset by annual benefits of \$72,000. By the end of the fifth year, the cumulative NPV reaches \$184,971, indicating strong financial viability and a favorable return on investment.

The analytical comparison further demonstrates that the proposed system has the potential to significantly reduce the time required for routine healthcare administrative tasks. Automated record management and digital appointment scheduling are expected to improve workflow efficiency and reduce staff workload, particularly in rural clinics where manual processes

remain prevalent. To further validate these findings, future work will focus on empirical performance evaluation and real-world system deployment.

The proposed system offers several advantages over the existing rural healthcare infrastructure, including faster digital patient registration, reduced data entry errors, searchable electronic records, efficient staff allocation, prevention of double booking, improved treatment accuracy through historical patient tracking, transparent billing processes, clear role-based user management, visual analytics to support decision-making, and robust data security ensured through role-based access control mechanisms.

The NPV and Return on Investment (ROI) calculations in this study are based on conservative assumptions regarding costs and benefits, reflecting the operational context of small to medium-sized rural healthcare clinics. The initial development cost includes software development and basic hardware setup, while recurring costs account for system maintenance and minimal cloud hosting fees. Anticipated benefits primarily arise from reduced administrative workload, decreased paper usage, and improved appointment scheduling efficiency.

Standard wage estimates for clinical support staff were used to convert healthcare workers' time savings into monetary values. A five-year evaluation period and a moderate discount rate were selected to account for long-term system adoption and operational sustainability. The assumptions underlying the analysis were informed by informal consultations with healthcare professionals and commonly reported pricing structures in rural healthcare environments. Rather than providing an exact financial forecast, this analysis aims to offer an indicative assessment of the system's economic feasibility.

TABLE II  
NET PRESENT VALUE (NPV) ANALYSIS

| Year | Initial Cost | Annual Benefit | Discount Factor | Present Value | Cumulative NPV |
|------|--------------|----------------|-----------------|---------------|----------------|
| 0    | 88,000       | 0              | 1.0000          | -88,000       | -88,000        |
| 1    | 0            | 72,000         | 0.9091          | 65,455        | -22,545        |
| 2    | 0            | 72,000         | 0.8264          | 59,507        | 36,962         |
| 3    | 0            | 72,000         | 0.7513          | 54,090        | 91,052         |
| 4    | 0            | 72,000         | 0.6830          | 49,176        | 140,228        |
| 5    | 0            | 72,000         | 0.6209          | 44,743        | 184,971        |

The analytical comparison indicates that the proposed system has the potential to substantially reduce the time required for routine healthcare administrative tasks. Automated record management and digital appointment scheduling are expected to improve workflow efficiency and reduce staff workload, particularly in rural clinics where manual processes remain prevalent. Future work will include real-world deployment and empirical performance measurement to further validate these findings.

The system demonstrates clear advantages over existing rural health systems through faster digital registration, reduced errors, searchable records, efficient staff assignment, prevention of double-booking, improved treatment accuracy through

historical tracking, transparent billing, clear role management, visual analytics for decision-making, and robust data protection through role-based access control Fig 4.

### C. Assumptions Used for Cost–Benefit Analysis

The study's Net Present Value (NPV) and Return on Investment (ROI) calculations are based on conservative cost and benefit assumptions appropriate for small to medium-sized rural healthcare clinics. Software development and basic hardware setup are included in the initial system development cost, however system maintenance and small cloud hosting charges are included in recurrent costs. Reductions in administrative workload, less paper use, and more effective appointment scheduling are the main sources of the predicted benefits. Standard wage estimates for clinical support staff were used to translate time savings for healthcare workers into monetary value. In order to account for long term system adoption and operational sustainability, a five year evaluation period and a moderate discount rate were considered. Informal conversations with healthcare professionals and pricing structures frequently reported in rural healthcare settings served as the basis for these presumptions. Rather than offering a precise financial forecast, the analysis is meant to offer an indicative evaluation of economic viability.

### D. Comparison with Existing Rural Health Information Systems

Comparing our system with other systems that are present in the rural health information domain we used two widely recognized health information systems and they are OpenMRS and DHIS2. Table III presents our findings where we made comparisons using key technical and operations parameters. As shown in Table III, existing systems such as OpenMRS and DHIS2 show great capabilities in healthcare data management; they do have setbacks such as proper infrastructure, technical expertise and stable internet connection. This is where our proposed system prioritizes offline operations to synchronize later with a simple deployment, SMS based patient alert. This is very useful for rural clinics to have access to limited resources.

TABLE III  
COMPARISON WITH EXISTING RURAL HIS

| Feature                      | Proposed System | OpenMRS        | DHIS2             |
|------------------------------|-----------------|----------------|-------------------|
| Offline Mode                 | Full            | Partial        | Limited           |
| SMS Notifications            | Yes             | No             | Limited           |
| Deployment Complexity        | Low             | Moderate       | High              |
| Infrastructure Requirement   | Minimal         | Moderate       | High              |
| Local Language Support       | Yes             | Partial        | Yes               |
| Technical Expertise Required | Low             | Moderate       | High              |
| Target User Scale            | Small Clinic    | Multi-facility | District/National |

## VIII. CONCLUSION

The Patient Information System for Rural Clinics is an improved tool for efficiency, decision-making, and care coordination by offering role-based access, offline functionality, reminder messages, and analytical processing. Developed by iteration and with community feedback, it resolves connectivity and human resource issues, is scalable, and manages public health with high ROI as well as equal access to health care for rural communities.

Medical Records

| Patient ID | Diagnosis    | Prescription  | Treatment Notes   |
|------------|--------------|---------------|-------------------|
| P001       | Fever        | Paracetamol   | Rest 3 days       |
| P002       | Diabetes     | Metformin     | Check sugar daily |
| P003       | Hypertension | Amlodipine    | Low salt diet     |
| P004       | Headache     | Ibuprofen     | Hydrate           |
| P005       | Fever        | Paracetamol   | Rest 2 days       |
| P006       | Cold         | Antihistamine | Warm fluids       |
| P007       | Diabetes     | Metformin     | Diet control      |
| P008       | Asthma       | Inhaler       | Avoid allergens   |
| P009       | Hypertension | Amlodipine    | Exercise daily    |
| P010       | Fever        | Paracetamol   | Rest 1 day        |

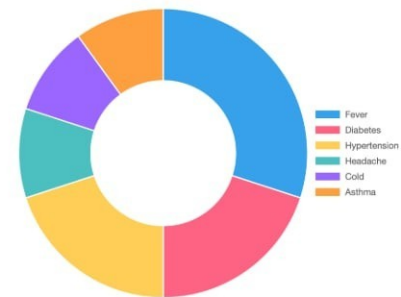


Fig. 4. Rich Picture

| Cumulative Costs (USD) | Cumulative Benefits (USD) | Net Cash Flow (USD) |
|------------------------|---------------------------|---------------------|
| -88,000                | 0                         | -88,000             |
| -88,000                | 65,455                    | -22,545             |
| -88,000                | 124,962                   | 36,962              |
| -88,000                | 179,052                   | 91,052              |
| -88,000                | 228,228                   | 140,228             |
| -88,000                | 272,971                   | 184,971             |

TABLE IV

RICH PICTURE

## REFERENCES

- [1] J. Braa, O. Hanseth, A. Heywood, W. Mohammed, and V. Shaw, "Developing health information systems in developing countries: The flexible standards strategy," *MIS Quarterly*, vol. 31, pp. 381–402, 2007.
- [2] V. M. D. Fava and L. V. Lapaaño, "Provision of digital primary health care services: Overview of reviews," *Journal of Medical Internet Research*, vol. 26, p. e53594, 2024.
- [3] H. Kabir, M. Hasan, N. Akter, U. Marma, T. Alam, A. Tutul, L. Biswas, R. Ara, and D. K. Mitra, "Factors associated with the intention of tele-health service utilization among bangladeshi people: A cross-sectional study," *F1000Research*, vol. 11, p. 996, 2022.



- [4] S. Davis, M. A. Smith, L. Burton, and K. L. Rush, "Personal health record implementation in rural primary care: A descriptive exploratory study using re-aim framework," *PLOS Digital Health*, vol. 3, no. 6, pp. 1–19, 2024.
- [5] M. Humayun, M. F. Almufareh, F. Al-Quayed, S. A. Alateyah, and M. Alatiyyah, "Improving healthcare facilities in remote areas using cutting-edge technologies," *Applied Sciences*, vol. 13, no. 11, 2023.
- [6] S. Baumann, R. T. Stone, and E. Abdelall, "Introducing a remote patient monitoring usability impact model to overcome challenges," *Sensors*, vol. 24, no. 12, 2024.
- [7] D. J. Smith, E. Mizelle, N. Ali, V. Cepeda, T. Pearson, K. Crumbley, D. Pimentel, S. H. Suarez, K. Mueller, Q. Phan, E. P. Ferranti, and L. A. Modly, "Improving rural healthcare in mobile clinics: Real-time, live data entry into the electronic medical record using a satellite internet connection," *International Journal of Environmental Research and Public Health*, vol. 22, no. 6, 2025.
- [8] J. Avrett, K. Bojan, K. Diep, H. Idrees, N. Mbunzama, K. Rajmohan, and A. Samineni, "Electronic health record architecture for rural clinic data connectivity and unique patient identification," 2013.
- [9] R. J. Mashoka, B. Murray, U. George, N. Lobue, J. Mfinanga, H. Sawe, and L. White, "Implementation of electronic medical records at an emergency medicine department in tanzania: The information technology perspective," *African Journal of Emergency Medicine*, vol. 9, no. 4, pp. 165–171, 2019.
- [10] L. Nguyen, E. Bellucci, and L. T. Nguyen, "Electronic health records implementation: An evaluation of information system impact and contingency factors," *International Journal of Medical Informatics*, vol. 83, no. 11, pp. 779–796, 2014.
- [11] J. Barjis, G. Kolfshoten, and J. Maritz, "A sustainable and affordable support system for rural healthcare delivery," *Decision Support Systems*, vol. 56, pp. 223–233, 2013.
- [12] S. J. Miah, J. Hasan, and J. G. Gammack, "On-cloud healthcare clinic: An e-health consultancy approach for remote communities in a developing country," *Telematics and Informatics*, vol. 34, no. 1, pp. 311–322, 2017.
- [13] I. Bardhan, R. Kohli, E. Oborn, A. Mishra, C. H. Tan, M. C. Tremblay, and S. Sarker, "Human-centric information systems research on the digital future of healthcare," *Information Systems Research*, vol. 36, no. 1, pp. 1–20, 2025.
- [14] P. Handayani, A. Hidayanto, and I. Budi, "User acceptance factors of hospital information systems and related technologies: Systematic review," *Informatics for Health and Social Care*, vol. 43, pp. 1–26, 2017.
- [15] R. Seethamraju, K. Diatha, and S. Garg, "Intention to use a mobile-based information technology solution for tuberculosis treatment monitoring – applying a utaut model," *Information Systems Frontiers*, vol. 20, 2018.
- [16] N. Hossain, F. Yokota, N. Sultana, and A. Ahmed, "Factors influencing rural end-users' acceptance of e-health in developing countries: A study on portable health clinic in bangladesh," *Telemedicine and e-Health*, vol. 25, no. 3, pp. 221–229, 2019.